

$$x = 0.082 \text{ m}$$

[A1]

- 6 (a) Straight lines in uniform radial pattern centred on charge [B1]

Arrows pointing inwards [B1]

- (b) Electric field points inwards, because E points from points of higher potential to lower potential.

OR potential is negative suggests that the sphere is negatively charged and hence electric field points inwards. [B1]

Electric field is numerically equal to the potential gradient and is stronger where the potential gradient is stronger (closer to the charged sphere). The stronger E field is shown by the closer spacing between the E field lines. [B1]

- (c) (i) If V is inversely proportional to x, then $Vx = \text{constant}$ [B1]

Multiplying **any 2** values [B1] to conclude that Vx is constant:

x / m	V / V	$Vx / \text{V m}$
0.19	-1.50×10^5	-28.5×10^3
0.25	-1.14×10^5	-28.5×10^3
0.32	-0.89×10^5	-28.48×10^3 $= -28.5 \times 10^3$
0.39	-0.73×10^5	-28.47×10^3 $= -28.5 \times 10^3$

- (ii) Using value of $V_x = -28.5 \times 10^3 \text{ V m}$ [M1]

$$R = -28.5 \times 10^3 / -1.9 \times 10^5 = 0.15 \text{ m} \text{ [A1]}$$

- (iii)

$$V = \frac{Q}{4\pi\epsilon_0 r} \text{ [C1]}$$

$$Q = 4\pi\epsilon_0 rV$$

$$= 4\pi(8.85 \times 10^{-12})(0.15)(-1.9 \times 10^5)$$

$$= -3.17 \times 10^{-6} \text{ C}$$

$$= -3.2 \times 10^{-6} \text{ C [A1]}$$